



## OVP Guide to Using Processor Models

### Model specific information for ARM\_Cortex-R8MPx3

Imperas Software Limited  
Imperas Buildings, North Weston  
Thame, Oxfordshire, OX9 2HA, U.K.  
[docs@imperas.com](mailto:docs@imperas.com)



|          |                                                      |
|----------|------------------------------------------------------|
| Author   | Imperas Software Limited                             |
| Version  | 20250716.0                                           |
| Filename | OVP_Model.Specific.Information_arm_Cortex-R8MPx3.pdf |
| Created  | 16 July 2025                                         |
| Status   | OVP Standard Release                                 |

## Copyright Notice

Copyright (c) 2024 Imperas Software Limited. All rights reserved. This software and documentation contain information that is the property of Imperas Software Limited. The software and documentation are furnished under a license agreement and may be used or copied only in accordance with the terms of the license agreement. No part of the software and documentation may be reproduced, transmitted, or translated, in any form or by any means, electronic, mechanical, manual, optical, or otherwise, without prior written permission of Imperas Software Limited, or as expressly provided by the license agreement.

## Right to Copy Documentation

The license agreement with Imperas permits licensee to make copies of the documentation for its internal use only. Each copy shall include all copyrights, trademarks, service marks, and proprietary rights notices, if any.

## Destination Control Statement

All technical data contained in this publication is subject to the export control laws of the United States of America. Disclosure to nationals of other countries contrary to United States law is prohibited. It is the readers responsibility to determine the applicable regulations and to comply with them.

## Disclaimer

IMPERAS SOFTWARE LIMITED, AND ITS LICENSORS MAKE NO WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, WITH REGARD TO THIS MATERIAL, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

## Model Release Status

This model is released as part of OVP releases and is included in OVPworld packages. Please visit [OVPworld.org](http://OVPworld.org).

# Contents

|          |                                                           |           |
|----------|-----------------------------------------------------------|-----------|
| <b>1</b> | <b>Overview</b>                                           | <b>1</b>  |
| 1.1      | Description . . . . .                                     | 1         |
| 1.2      | Licensing . . . . .                                       | 1         |
| 1.3      | Limitations . . . . .                                     | 2         |
| 1.4      | Verification . . . . .                                    | 2         |
| 1.5      | Features . . . . .                                        | 3         |
| 1.5.1    | Core Features . . . . .                                   | 3         |
| 1.5.2    | Memory System . . . . .                                   | 3         |
| 1.5.3    | Advanced SIMD and Floating-Point Features . . . . .       | 3         |
| 1.5.4    | Generic Interrupt Controller . . . . .                    | 3         |
| 1.6      | Debug Mask . . . . .                                      | 4         |
| 1.7      | AArch32 Unpredictable Behavior . . . . .                  | 4         |
| 1.7.1    | Equal Target Registers . . . . .                          | 4         |
| 1.7.2    | Floating Point Load/Store Multiple Lists . . . . .        | 4         |
| 1.7.3    | Floating Point VLD[2-4]/VST[2-4] Range Overflow . . . . . | 5         |
| 1.7.4    | If-Then (IT) Block Constraints . . . . .                  | 5         |
| 1.7.5    | Use of R13 . . . . .                                      | 5         |
| 1.7.6    | Use of R15 . . . . .                                      | 5         |
| 1.7.7    | Unpredictable Instructions in Some Modes . . . . .        | 5         |
| 1.8      | Integration Support . . . . .                             | 6         |
| 1.8.1    | Halt Reason Introspection . . . . .                       | 6         |
| 1.8.2    | Automatic WFI Wake Up . . . . .                           | 6         |
| 1.8.3    | System Register Access Monitor . . . . .                  | 6         |
| 1.8.4    | System Register Implementation . . . . .                  | 7         |
| <b>2</b> | <b>Configuration</b>                                      | <b>8</b>  |
| 2.1      | Location . . . . .                                        | 8         |
| 2.2      | GDB Path . . . . .                                        | 8         |
| 2.3      | Semi-Host Library . . . . .                               | 8         |
| 2.4      | Processor Endian-ness . . . . .                           | 8         |
| 2.5      | QuantumLeap Support . . . . .                             | 8         |
| 2.6      | Processor ELF code . . . . .                              | 8         |
| <b>3</b> | <b>All Variants in this model</b>                         | <b>9</b>  |
| <b>4</b> | <b>Bus Master Ports</b>                                   | <b>13</b> |
| <b>5</b> | <b>Bus Slave Ports</b>                                    | <b>14</b> |

|           |                                       |           |
|-----------|---------------------------------------|-----------|
| <b>6</b>  | <b>Net Ports</b>                      | <b>15</b> |
| <b>7</b>  | <b>FIFO Ports</b>                     | <b>20</b> |
| <b>8</b>  | <b>Formal Parameters</b>              | <b>21</b> |
| 8.1       | Parameter values and limits . . . . . | 26        |
| <b>9</b>  | <b>Execution Modes</b>                | <b>30</b> |
| <b>10</b> | <b>Exceptions</b>                     | <b>31</b> |
| <b>11</b> | <b>Hierarchy of the model</b>         | <b>32</b> |
| 11.1      | Level 1: MPCORE . . . . .             | 32        |
| 11.2      | Level 2: CPU . . . . .                | 32        |
| <b>12</b> | <b>Model Commands</b>                 | <b>34</b> |
| 12.1      | Level 1: MPCORE . . . . .             | 34        |
| 12.1.1    | isync . . . . .                       | 34        |
| 12.1.2    | itrace . . . . .                      | 34        |
| 12.2      | Level 2: CPU . . . . .                | 35        |
| 12.2.1    | debugflags . . . . .                  | 35        |
| 12.2.2    | isync . . . . .                       | 35        |
| 12.2.3    | itrace . . . . .                      | 35        |
| 12.2.4    | listSysRegsAA32 . . . . .             | 36        |
| 12.2.4.1  | Argument description . . . . .        | 36        |
| <b>13</b> | <b>Registers</b>                      | <b>37</b> |
| 13.1      | Level 1: MPCORE . . . . .             | 37        |
| 13.2      | Level 2: CPU . . . . .                | 37        |
| 13.2.1    | Core . . . . .                        | 37        |
| 13.2.2    | Control . . . . .                     | 37        |
| 13.2.3    | User . . . . .                        | 38        |
| 13.2.4    | FIQ . . . . .                         | 38        |
| 13.2.5    | IRQ . . . . .                         | 38        |
| 13.2.6    | Supervisor . . . . .                  | 38        |
| 13.2.7    | Undefined . . . . .                   | 39        |
| 13.2.8    | Abort . . . . .                       | 39        |
| 13.2.9    | SIMD_VFP . . . . .                    | 39        |
| 13.2.10   | SIMD_VFP_SYS . . . . .                | 39        |
| 13.2.11   | Coprocessor_32_bit . . . . .          | 40        |
| 13.2.12   | Integration_support . . . . .         | 41        |
| 13.2.13   | MPCore_SCR . . . . .                  | 41        |
| 13.2.14   | MPCore_distributor . . . . .          | 42        |
| 13.2.15   | MPCore_processor_interface . . . . .  | 44        |
| 13.2.16   | MPCore_timer_and_watchdog . . . . .   | 44        |
| 13.2.17   | MPCore_global_timer . . . . .         | 44        |

# Chapter 1

## Overview

This document provides the details of an OVP Fast Processor Model variant.

OVP Fast Processor Models are written in C and provide a C API for use in C based platforms. The models also provide a native interface for use in SystemC TLM2 platforms.

The models are written using the OVP VMI API that provides a Virtual Machine Interface that defines the behavior of the processor. The VMI API makes a clear line between model and simulator allowing very good optimization and world class high speed performance. Most models are provided as a binary shared object and also as source. This allows the download and use of the model binary or the use of the source to explore and modify the model.

The models are run through an extensive QA and regression testing process and most model families are validated using technology provided by the processor IP owners. There is a companion document (OVP Guide to Using Processor Models) which explains the general concepts of OVP Fast Processor Models and their use. It is downloadable from the OVPworld website documentation pages.

### 1.1 Description

ARM Processor Model

### 1.2 Licensing

Usage of binary model under license governing simulator usage.

Note that for models of ARM CPUs the license includes the following terms:

Licensee is granted a non-exclusive, worldwide, non-transferable, revocable licence to:

If no source is being provided to the Licensee: use and copy only (no modifications rights are granted) the model for the sole purpose of designing, developing, analyzing, debugging, testing, verifying, validating and optimizing software which: (a) (i) is for ARM based systems; and (ii) does not incorporate the ARM Models or any part thereof; and (b) such ARM Models may not be used

to emulate an ARM based system to run application software in a production or live environment.

If source code is being provided to the Licensee: use, copy and modify the model for the sole purpose of designing, developing, analyzing, debugging, testing, verifying, validating and optimizing software which: (a) (i) is for ARM based systems; and (ii) does not incorporate the ARM Models or any part thereof; and (b) such ARM Models may not be used to emulate an ARM based system to run application software in a production or live environment.

In the case of any Licensee who is either or both an academic or educational institution the purposes shall be limited to internal use.

Except to the extent that such activity is permitted by applicable law, Licensee shall not reverse engineer, decompile, or disassemble this model. If this model was provided to Licensee in Europe, Licensee shall not reverse engineer, decompile or disassemble the Model for the purposes of error correction.

The License agreement does not entitle Licensee to manufacture in silicon any product based on this model.

The License agreement does not entitle Licensee to use this model for evaluating the validity of any ARM patent.

Source of model available under separate Imperas Software License Agreement.

## 1.3 Limitations

Instruction pipelines are not modeled in any way. All instructions are assumed to complete immediately. This means that instruction barrier instructions (e.g. ISB, CP15ISB) are treated as NOPs, with the exception of any undefined instruction behavior, which is modeled. The model does not implement speculative fetch behavior. The branch cache is not modeled.

Caches and write buffers are not modeled in any way. All loads, fetches and stores complete immediately and in order, and are fully synchronous (as if the memory was of Strongly Ordered or Device-nGnRnE type). Data barrier instructions (e.g. DSB, CP15DSB) are treated as NOPs, with the exception of any undefined instruction behavior, which is modeled. Cache manipulation instructions are implemented as NOPs, with the exception of any undefined instruction behavior, which is modeled.

Real-world timing effects are not modeled: all instructions are assumed to complete in a single cycle.

Performance Monitors are implemented as a register interface only except for the cycle counter, which is implemented assuming one instruction per cycle.

## 1.4 Verification

Models have been extensively tested by Imperas. ARM Cortex models have been successfully used by customers to simulate SMP Linux, Ubuntu Desktop, VxWorks and ThreadX on Xilinx Zynq virtual platforms.

## 1.5 Features

The precise set of implemented features in the model is defined by ID registers. Use overrides to modify these if required (for example `override_PFR0` or `override_AA64PFR0_EL1`).

### 1.5.1 Core Features

Thumb-2 instructions are supported.

Trivial Jazelle extension is implemented.

### 1.5.2 Memory System

FCSE extension is implemented.

PMSA address translation is implemented with a unified MPU with 12 regions. Use parameter “`override_MPUIR`” to specify a different MPU size if required.

1 ITCM is implemented.

1 DTCM is implemented.

For each TCM, a corresponding bus port is present. If parameter “`useInternalTCMs`” is `True`, TCMs are modelled internally and the port is a target port, allowing TCM contents to be examined and updated externally. Otherwise, if parameter “`useInternalTCMs`” is `False`, the corresponding port is an initiator port to which an external TCM memory model must be connected.

### 1.5.3 Advanced SIMD and Floating-Point Features

VFP instructions are implemented.

The model implements trapped exceptions if `FPTrap` is set to 1 in `MVFR0` (for AArch32) or `MVFR0_EL1` (for AArch64). When floating point exception traps are taken, cumulative exception flags are not updated (in other words, cumulative flag state is always the same as prior to instruction execution, even for SIMD instructions). When multiple enabled exceptions are raised by a single floating point operation, the exception reported is the one in least-significant bit position in `FPSCR` (for AArch32) or `FPCR` (for AArch64). When multiple enabled exceptions are raised by different SIMD element computations, the exception reported is selected from the lowest-index-number SIMD operation. Contact Imperas if requirements for exception reporting differ from these.

Trapped exceptions not are implemented in this variant (`FPTrap=0`)

### 1.5.4 Generic Interrupt Controller

GIC block is implemented (GICv1). Accesses to GIC registers can be viewed externally by connecting to the 32-bit `GICRegisters` bus port. Secure register accesses are at offset 0x0 on this bus; for example, a secure access to GIC register `ICDDCR` can be observed by monitoring address

0x00001000. Non-secure accesses are at offset 0x80000000 on this bus; for example, a non-secure access to GIC register ICDDCR can be observed by monitoring address 0x80001000

## 1.6 Debug Mask

It is possible to enable model debug features in various categories. This can be done statically using the “override\_debugMask” parameter, or dynamically using the “debugflags” command. Enabled debug features are specified using a bitmask value, as follows:

Value 0x004: enable debugging of MMU/MPU mappings.

Value 0x020: enable debugging of reads and writes of GIC block registers.

Value 0x040: enable debugging of exception routing via the GIC model component.

Value 0x080: enable debugging of all system register accesses.

Value 0x100: enable debugging of all traps of system register accesses.

Value 0x200: enable verbose debugging of other miscellaneous behavior (for example, the reason why a particular instruction is undefined).

Value 0x400: enable debugging of Performance Monitor timers

All other bits in the debug bitmask are reserved and must not be set to non-zero values.

## 1.7 AArch32 Unpredictable Behavior

Many AArch32 instruction behaviors are described in the ARM ARM as CONSTRAINED UNPREDICTABLE. This section describes how such situations are handled by this model.

### 1.7.1 Equal Target Registers

Some instructions allow the specification of two target registers (for example, double-width SMULL, or some VMOV variants), and such instructions are CONSTRAINED UNPREDICTABLE if the same target register is specified in both positions. In this model, such instructions are treated as UNDEFINED.

### 1.7.2 Floating Point Load/Store Multiple Lists

Instructions that load or store a list of floating point registers (e.g. VSTM, VLDM, VPUSH, VPOP) are CONSTRAINED UNPREDICTABLE if either the uppermost register in the specified range is greater than 32 or (for 64-bit registers) if more than 16 registers are specified. In this model, such instructions are treated as UNDEFINED.



### 1.7.3 Floating Point VLD[2-4]/VST[2-4] Range Overflow

Instructions that load or store a fixed number of floating point registers (e.g. VST2, VLD2) are CONSTRAINED UNPREDICTABLE if the upper register bound exceeds the number of implemented floating point registers. In this model, these instructions load and store using modulo 32 indexing (consistent with AArch64 instructions with similar behavior).

### 1.7.4 If-Then (IT) Block Constraints

Where the behavior of an instruction in an if-then (IT) block is described as CONSTRAINED UNPREDICTABLE, this model treats that instruction as UNDEFINED.

### 1.7.5 Use of R13

In architecture variants before ARMv8, use of R13 was described as CONSTRAINED UNPREDICTABLE in many circumstances. From ARMv8, most of these situations are no longer considered unpredictable. This model allows R13 to be used like any other GPR, consistent with the ARMv8 specification.

### 1.7.6 Use of R15

Use of R15 is described as CONSTRAINED UNPREDICTABLE in many circumstances. This model allows such use to be configured using the parameter “unpredictableR15” as follows:

Value “undefined”: any reference to R15 in such a situation is treated as UNDEFINED;

Value “nop”: any reference to R15 in such a situation causes the instruction to be treated as a NOP;

Value “raz\_wi”: any reference to R15 in such a situation causes the instruction to be treated as a RAZ/WI (that is, R15 is read as zero and write-ignored);

Value “execute”: any reference to R15 in such a situation is executed using the current value of R15 on read, and writes to R15 are allowed (but are not interworking).

Value “assert”: any reference to R15 in such a situation causes the simulation to halt with an assertion message (allowing any such unpredictable uses to be easily identified).

In this variant, the default value of “unpredictableR15” is “undefined”.

### 1.7.7 Unpredictable Instructions in Some Modes

Some instructions are described as CONSTRAINED UNPREDICTABLE in some modes only (for example, MSR accessing SPSR is CONSTRAINED UNPREDICTABLE in User and System modes). This model allows such use to be configured using the parameter “unpredictableModal”, which can have values “undefined” or “nop”. See the previous section for more information about the meaning of these values.

In this variant, the default value of “unpredictableModal” is “nop”.

## 1.8 Integration Support

This model implements a number of non-architectural pseudo-registers and other features to facilitate integration.

### 1.8.1 Halt Reason Introspection

An artifact register HaltReason can be read to determine the reason or reasons that a processor is halted. This register is a bitfield, with the following encoding: bit 0 indicates the processor has executed a wait-for-event (WFE) instruction; bit 1 indicates the processor has executed a wait-for-interrupt (WFI) instruction; and bit 2 indicates the processor is held in reset.

### 1.8.2 Automatic WFI Wake Up

A simulation environment can cause a core to wake up from the WFI state by writing to the artifact register WakeWFINow.

### 1.8.3 System Register Access Monitor

If parameter “enableSystemMonitorBus” is True, an artifact 32-bit bus “SystemMonitor” is enabled for each PE. Every system register read or write by that PE is then visible as a read or write on this artifact bus, and can therefore be monitored using callbacks installed in the client environment (use opBusReadMonitorAdd/opBusWriteMonitorAdd or icmAddBusReadCallback/icmAddBusWriteCallback, depending on the client API). The format of the address on the bus is as follows:

bits 31:26 - zero

bit 25 - 1 if AArch64 access, 0 if AArch32 access

bit 24 - 1 if non-secure access, 0 if secure access

bits 23:20 - CRm value

bits 19:16 - CRn value

bits 15:12 - op2 value

bits 11:8 - op1 value

bits 7:4 - op0 value (AArch64) or coprocessor number (AArch32)

bits 3:0 - zero

As an example, to view non-secure writes to writes to CNTFRQ\_EL0 in AArch64 state, install a write monitor on address range 0x020e0330:0x020e0333.

#### 1.8.4 System Register Implementation

If parameter “enableSystemBus” is True, an artifact 32-bit bus “System” is enabled for each PE. Slave callbacks installed on this bus can be used to implement modified system register behavior (use opBusSlaveNew or icmMapExternalMemory, depending on the client API). The format of the address on the bus is the same as for the system monitor bus, described above.

## Chapter 2

# Configuration

### 2.1 Location

This model's VLVN is `arm.ovpworld.org/processor/arm/1.0`.

The model source is usually at:

`$IMPERAS_HOME/ImperasLib/source/arm.ovpworld.org/processor/arm/1.0`

The model binary is usually at:

`$IMPERAS_HOME/lib/$IMPERAS_ARCH/ImperasLib/arm.ovpworld.org/processor/arm/1.0`

### 2.2 GDB Path

The default GDB for this model is: `$IMPERAS_HOME/lib/$IMPERAS_ARCH/gdb/arm-none-eabi-gdb`.

### 2.3 Semi-Host Library

The default semi-host library file is `arm.ovpworld.org/semihosting/armNewlib/1.0`

### 2.4 Processor Endian-ness

This is a LITTLE endian model.

### 2.5 QuantumLeap Support

This processor is qualified to run in a QuantumLeap enabled simulator.

### 2.6 Processor ELF code

The ELF code supported by this model is: `0x28`.

## Chapter 3

# All Variants in this model

This model has these variants

| Variant     | Description |
|-------------|-------------|
| ARMv4T      |             |
| ARMv4xM     |             |
| ARMv4       |             |
| ARMv4TxM    |             |
| ARMv5xM     |             |
| ARMv5       |             |
| ARMv5TxM    |             |
| ARMv5T      |             |
| ARMv5TExP   |             |
| ARMv5TE     |             |
| ARMv5TEJ    |             |
| ARMv6       |             |
| ARMv6K      |             |
| ARMv6T2     |             |
| ARMv6KZ     |             |
| ARMv7       |             |
| ARM7TDMI    |             |
| ARM7EJ-S    |             |
| ARM720T     |             |
| ARM920T     |             |
| ARM922T     |             |
| ARM926EJ-S  |             |
| ARM940T     |             |
| ARM946E     |             |
| ARM966E     |             |
| ARM968E-S   |             |
| ARM1020E    |             |
| ARM1022E    |             |
| ARM1026EJ-S |             |
| ARM1136J-S  |             |
| ARM1156T2-S |             |

|                 |                              |
|-----------------|------------------------------|
| ARM1176JZ-S     |                              |
| Cortex-R4       |                              |
| Cortex-R4F      |                              |
| Cortex-R5MPx1   |                              |
| Cortex-R5MPx2   |                              |
| Cortex-R8MPx1   |                              |
| Cortex-R8MPx2   |                              |
| Cortex-R8MPx3   | (described in this document) |
| Cortex-R8MPx4   |                              |
| Cortex-R52MPx1  |                              |
| Cortex-R52MPx2  |                              |
| Cortex-R52MPx3  |                              |
| Cortex-R52MPx4  |                              |
| Cortex-R52+MPx1 |                              |
| Cortex-R52+MPx2 |                              |
| Cortex-R52+MPx3 |                              |
| Cortex-R52+MPx4 |                              |
| Cortex-R82MPx1  |                              |
| Cortex-R82MPx2  |                              |
| Cortex-R82MPx3  |                              |
| Cortex-R82MPx4  |                              |
| Cortex-R82MPx5  |                              |
| Cortex-R82MPx6  |                              |
| Cortex-R82MPx7  |                              |
| Cortex-R82MPx8  |                              |
| Cortex-A5UP     |                              |
| Cortex-A5MPx1   |                              |
| Cortex-A5MPx2   |                              |
| Cortex-A5MPx3   |                              |
| Cortex-A5MPx4   |                              |
| Cortex-A8       |                              |
| Cortex-A9UP     |                              |
| Cortex-A9MPx1   |                              |
| Cortex-A9MPx2   |                              |
| Cortex-A9MPx3   |                              |
| Cortex-A9MPx4   |                              |
| Cortex-A7UP     |                              |
| Cortex-A7MPx1   |                              |
| Cortex-A7MPx2   |                              |
| Cortex-A7MPx3   |                              |
| Cortex-A7MPx4   |                              |
| Cortex-A15UP    |                              |
| Cortex-A15MPx1  |                              |
| Cortex-A15MPx2  |                              |
| Cortex-A15MPx3  |                              |

|                  |  |
|------------------|--|
| Cortex-A15MPx4   |  |
| Cortex-A17MPx1   |  |
| Cortex-A17MPx2   |  |
| Cortex-A17MPx3   |  |
| Cortex-A17MPx4   |  |
| AArch32          |  |
| AArch64          |  |
| Cortex-A32MPx1   |  |
| Cortex-A32MPx2   |  |
| Cortex-A32MPx3   |  |
| Cortex-A32MPx4   |  |
| Cortex-A35MPx1   |  |
| Cortex-A35MPx2   |  |
| Cortex-A35MPx3   |  |
| Cortex-A35MPx4   |  |
| Cortex-A53MPx1   |  |
| Cortex-A53MPx2   |  |
| Cortex-A53MPx3   |  |
| Cortex-A53MPx4   |  |
| Cortex-A55MPx1   |  |
| Cortex-A55MPx2   |  |
| Cortex-A55MPx3   |  |
| Cortex-A55MPx4   |  |
| Cortex-A57MPx1   |  |
| Cortex-A57MPx2   |  |
| Cortex-A57MPx3   |  |
| Cortex-A57MPx4   |  |
| Cortex-A65AEMPx1 |  |
| Cortex-A65AEMPx2 |  |
| Cortex-A65AEMPx3 |  |
| Cortex-A65AEMPx4 |  |
| Cortex-A72MPx1   |  |
| Cortex-A72MPx2   |  |
| Cortex-A72MPx3   |  |
| Cortex-A72MPx4   |  |
| Cortex-A73MPx1   |  |
| Cortex-A73MPx2   |  |
| Cortex-A73MPx3   |  |
| Cortex-A73MPx4   |  |
| Cortex-A75MPx1   |  |
| Cortex-A75MPx2   |  |
| Cortex-A75MPx3   |  |
| Cortex-A75MPx4   |  |
| Cortex-A76MPx1   |  |
| Cortex-A76MPx2   |  |

|                |  |
|----------------|--|
| Cortex-A76MPx3 |  |
| Cortex-A76MPx4 |  |
| Cortex-A78MPx1 |  |
| Cortex-A78MPx2 |  |
| Cortex-A78MPx3 |  |
| Cortex-A78MPx4 |  |
| MultiCluster   |  |

Table 3.1: All Variants in this model



## Chapter 4

# Bus Master Ports

This model has these bus master ports.

| Name         | min | max | Connect?  | Description                      |
|--------------|-----|-----|-----------|----------------------------------|
| ITCM0_CPU0   | 13  | 13  | optional  | instruction TCM                  |
| DTCM0_CPU0   | 13  | 13  | optional  | data TCM                         |
| FPP_CPU0     | 32  | 32  | optional  | Fast peripheral port             |
| LLPP         | 32  | 32  | optional  | Low-latency peripheral port      |
| ITCM0_CPU1   | 13  | 13  | optional  | instruction TCM                  |
| DTCM0_CPU1   | 13  | 13  | optional  | data TCM                         |
| FPP_CPU1     | 32  | 32  | optional  | Fast peripheral port             |
| ITCM0_CPU2   | 13  | 13  | optional  | instruction TCM                  |
| DTCM0_CPU2   | 13  | 13  | optional  | data TCM                         |
| FPP_CPU2     | 32  | 32  | optional  | Fast peripheral port             |
| INSTRUCTION  | 32  | 32  | mandatory |                                  |
| DATA         | 32  | 32  | optional  |                                  |
| GICRegisters | 32  | 32  | optional  | GIC memory-mapped register block |

Table 4.1: Bus Master Ports

## Chapter 5

# Bus Slave Ports

This model has no bus slave ports.

## Chapter 6

# Net Ports

This model has these net ports.

| Name  | Type  | Connect? | Description                 |
|-------|-------|----------|-----------------------------|
| SPI32 | input | optional | Shared peripheral interrupt |
| SPI33 | input | optional | Shared peripheral interrupt |
| SPI34 | input | optional | Shared peripheral interrupt |
| SPI35 | input | optional | Shared peripheral interrupt |
| SPI36 | input | optional | Shared peripheral interrupt |
| SPI37 | input | optional | Shared peripheral interrupt |
| SPI38 | input | optional | Shared peripheral interrupt |
| SPI39 | input | optional | Shared peripheral interrupt |
| SPI40 | input | optional | Shared peripheral interrupt |
| SPI41 | input | optional | Shared peripheral interrupt |
| SPI42 | input | optional | Shared peripheral interrupt |
| SPI43 | input | optional | Shared peripheral interrupt |
| SPI44 | input | optional | Shared peripheral interrupt |
| SPI45 | input | optional | Shared peripheral interrupt |
| SPI46 | input | optional | Shared peripheral interrupt |
| SPI47 | input | optional | Shared peripheral interrupt |
| SPI48 | input | optional | Shared peripheral interrupt |
| SPI49 | input | optional | Shared peripheral interrupt |
| SPI50 | input | optional | Shared peripheral interrupt |
| SPI51 | input | optional | Shared peripheral interrupt |
| SPI52 | input | optional | Shared peripheral interrupt |
| SPI53 | input | optional | Shared peripheral interrupt |
| SPI54 | input | optional | Shared peripheral interrupt |
| SPI55 | input | optional | Shared peripheral interrupt |
| SPI56 | input | optional | Shared peripheral interrupt |
| SPI57 | input | optional | Shared peripheral interrupt |
| SPI58 | input | optional | Shared peripheral interrupt |
| SPI59 | input | optional | Shared peripheral interrupt |
| SPI60 | input | optional | Shared peripheral interrupt |
| SPI61 | input | optional | Shared peripheral interrupt |

|                 |        |          |                                                      |
|-----------------|--------|----------|------------------------------------------------------|
| SPI62           | input  | optional | Shared peripheral interrupt                          |
| SPI63           | input  | optional | Shared peripheral interrupt                          |
| SPI64           | input  | optional | Shared peripheral interrupt                          |
| SPI65           | input  | optional | Shared peripheral interrupt                          |
| SPI66           | input  | optional | Shared peripheral interrupt                          |
| SPI67           | input  | optional | Shared peripheral interrupt                          |
| SPI68           | input  | optional | Shared peripheral interrupt                          |
| SPI69           | input  | optional | Shared peripheral interrupt                          |
| SPI70           | input  | optional | Shared peripheral interrupt                          |
| SPI71           | input  | optional | Shared peripheral interrupt                          |
| SPI72           | input  | optional | Shared peripheral interrupt                          |
| SPI73           | input  | optional | Shared peripheral interrupt                          |
| SPI74           | input  | optional | Shared peripheral interrupt                          |
| SPI75           | input  | optional | Shared peripheral interrupt                          |
| SPI76           | input  | optional | Shared peripheral interrupt                          |
| SPI77           | input  | optional | Shared peripheral interrupt                          |
| SPI78           | input  | optional | Shared peripheral interrupt                          |
| SPI79           | input  | optional | Shared peripheral interrupt                          |
| SPI80           | input  | optional | Shared peripheral interrupt                          |
| SPI81           | input  | optional | Shared peripheral interrupt                          |
| SPI82           | input  | optional | Shared peripheral interrupt                          |
| SPI83           | input  | optional | Shared peripheral interrupt                          |
| SPI84           | input  | optional | Shared peripheral interrupt                          |
| SPI85           | input  | optional | Shared peripheral interrupt                          |
| SPI86           | input  | optional | Shared peripheral interrupt                          |
| SPI87           | input  | optional | Shared peripheral interrupt                          |
| SPI88           | input  | optional | Shared peripheral interrupt                          |
| SPI89           | input  | optional | Shared peripheral interrupt                          |
| SPI90           | input  | optional | Shared peripheral interrupt                          |
| SPI91           | input  | optional | Shared peripheral interrupt                          |
| SPI92           | input  | optional | Shared peripheral interrupt                          |
| SPI93           | input  | optional | Shared peripheral interrupt                          |
| SPI94           | input  | optional | Shared peripheral interrupt                          |
| SPI95           | input  | optional | Shared peripheral interrupt                          |
| SPIVector       | input  | optional | Shared peripheral interrupt vectorized input         |
| periphReset     | input  | optional | Peripheral reset (active high)                       |
| EVENTI          | input  | optional | Event input signal, active on rising edge            |
| EVENTO          | output | optional | Event output signal, active on rising edge           |
| wdResetReq_CPU0 | output | optional | Watchdog interrupt request                           |
| wdReset_CPU0    | input  | optional | Watchdog reset (active high)                         |
| scuReset        | input  | optional | SCU reset (active high)                              |
| MFILTEREN       | input  | optional | Configure Address Filtering enable bit (active high) |
| MFILTERSTART    | input  | optional | Configure AF start address for AXIM1                 |

|                  |        |          |                                                                             |
|------------------|--------|----------|-----------------------------------------------------------------------------|
| MFILTEREND       | input  | optional | Configure AF end address for AXIM1                                          |
| PFILTERSTART     | input  | optional | Configure AF start address for AXIM1                                        |
| PFILTEREND       | input  | optional | Configure AF end address for AXIM1                                          |
| FPPSTART0        | input  | optional | Configure AF start address for FPP on core 0                                |
| FPPEND0          | input  | optional | Configure AF end address for FPP on core 0                                  |
| FPPSTART1        | input  | optional | Configure AF start address for FPP on core 1                                |
| FPPEND1          | input  | optional | Configure AF end address for FPP on core 1                                  |
| FPPSTART2        | input  | optional | Configure AF start address for FPP on core 2                                |
| FPPEND2          | input  | optional | Configure AF end address for FPP on core 2                                  |
| FPPSTART3        | input  | optional | Configure AF start address for FPP on core 3                                |
| FPPEND3          | input  | optional | Configure AF end address for FPP on core 3                                  |
| IRQOUT_CPU0      | output | optional | IRQ wakeup                                                                  |
| FIQOUT_CPU0      | output | optional | FIQ wakeup                                                                  |
| CLUSTERIDAFF1    | input  | optional | Configure MPIDR.Aff1                                                        |
| CLUSTERIDAFF2    | input  | optional | Configure MPIDR.Aff2                                                        |
| PERIPHBASE       | input  | optional | Base address for timers, watchdogs, interrupt controller, and SCU registers |
| VINITHI_CPU0     | input  | optional | Configure HIVECS mode (SCTLR.V)                                             |
| CFGEND_CPU0      | input  | optional | Configure exception endianness (SCTLR.EE)                                   |
| INITRAMI_CPU0    | input  | optional | Configure ITCM enable bit (active high)                                     |
| TEINIT_CPU0      | input  | optional | Configure exception state at reset (SCTLR.TE)                               |
| CFGNMFI_CPU0     | input  | optional | Configure non-maskable fast interrupts (SCTLR.NMFI)                         |
| reset_CPU0       | input  | optional | Processor reset, active high                                                |
| fiq_CPU0         | input  | optional | FIQ interrupt, active high (negation of nFIQ)                               |
| irq_CPU0         | input  | optional | IRQ interrupt, active high (negation of nIRQ)                               |
| sei_CPU0         | input  | optional | System error interrupt, active on rising edge (negation of nSEI)            |
| cpuHalt_CPU0     | input  | optional | Halt on reset                                                               |
| haltReason_CPU0  | output | optional | Indicates why core is halted                                                |
| AXI_SLVERR_CPU0  | input  | optional | AXI external abort type (DECERR=0, SLVERR=1)                                |
| syncCluster_CPU0 | output | optional | Cluster synchronization required                                            |
| updateTimer_CPU0 | output | optional | Internal timer update                                                       |

|                  |        |          |                                                                  |
|------------------|--------|----------|------------------------------------------------------------------|
| FPUFLAGS_CPU0    | output | optional | Cumulative exception flags signal output                         |
| SMPnAMP_CPU0     | output | optional | SMP or AMP mode                                                  |
| wdResetReq_CPU1  | output | optional | Watchdog interrupt request                                       |
| wdReset_CPU1     | input  | optional | Watchdog reset (active high)                                     |
| IRQOUT_CPU1      | output | optional | IRQ wakeup                                                       |
| FIQOUT_CPU1      | output | optional | FIQ wakeup                                                       |
| VINITHI_CPU1     | input  | optional | Configure HIVECS mode (SCTLR.V)                                  |
| CFGEND_CPU1      | input  | optional | Configure exception endianness (SCTLR.EE)                        |
| INTRAML_CPU1     | input  | optional | Configure ITCM enable bit (active high)                          |
| TEINIT_CPU1      | input  | optional | Configure exception state at reset (SCTLR.TE)                    |
| CFGNMFI_CPU1     | input  | optional | Configure non-maskable fast interrupts (SCTLR.NMFI)              |
| reset_CPU1       | input  | optional | Processor reset, active high                                     |
| fiq_CPU1         | input  | optional | FIQ interrupt, active high (negation of nFIQ)                    |
| irq_CPU1         | input  | optional | IRQ interrupt, active high (negation of nIRQ)                    |
| sei_CPU1         | input  | optional | System error interrupt, active on rising edge (negation of nSEI) |
| cpuHalt_CPU1     | input  | optional | Halt on reset                                                    |
| haltReason_CPU1  | output | optional | Indicates why core is halted                                     |
| AXI_SLVERR_CPU1  | input  | optional | AXI external abort type (DECERR=0, SLVERR=1)                     |
| syncCluster_CPU1 | output | optional | Cluster synchronization required                                 |
| updateTimer_CPU1 | output | optional | Internal timer update                                            |
| FPUFLAGS_CPU1    | output | optional | Cumulative exception flags signal output                         |
| SMPnAMP_CPU1     | output | optional | SMP or AMP mode                                                  |
| wdResetReq_CPU2  | output | optional | Watchdog interrupt request                                       |
| wdReset_CPU2     | input  | optional | Watchdog reset (active high)                                     |
| IRQOUT_CPU2      | output | optional | IRQ wakeup                                                       |
| FIQOUT_CPU2      | output | optional | FIQ wakeup                                                       |
| VINITHI_CPU2     | input  | optional | Configure HIVECS mode (SCTLR.V)                                  |
| CFGEND_CPU2      | input  | optional | Configure exception endianness (SCTLR.EE)                        |
| INTRAML_CPU2     | input  | optional | Configure ITCM enable bit (active high)                          |
| TEINIT_CPU2      | input  | optional | Configure exception state at reset (SCTLR.TE)                    |
| CFGNMFI_CPU2     | input  | optional | Configure non-maskable fast interrupts (SCTLR.NMFI)              |
| reset_CPU2       | input  | optional | Processor reset, active high                                     |
| fiq_CPU2         | input  | optional | FIQ interrupt, active high (negation of nFIQ)                    |
| irq_CPU2         | input  | optional | IRQ interrupt, active high (negation of nIRQ)                    |

|                  |        |          |                                                                  |
|------------------|--------|----------|------------------------------------------------------------------|
| sei_CPU2         | input  | optional | System error interrupt, active on rising edge (negation of nSEI) |
| cpuHalt_CPU2     | input  | optional | Halt on reset                                                    |
| haltReason_CPU2  | output | optional | Indicates why core is halted                                     |
| AXI_SLVERR_CPU2  | input  | optional | AXI external abort type (DECERR=0, SLVERR=1)                     |
| syncCluster_CPU2 | output | optional | Cluster synchronization required                                 |
| updateTimer_CPU2 | output | optional | Internal timer update                                            |
| FPUFLAGS_CPU2    | output | optional | Cumulative exception flags signal output                         |
| SMPnAMP_CPU2     | output | optional | SMP or AMP mode                                                  |

Table 6.1: Net Ports

## Chapter 7

# FIFO Ports

This model has no FIFO ports.



## Chapter 8

# Formal Parameters

| Name                   | Type        | Description                                                                                                                                                               |
|------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| verbose                | Boolean     | Specify verbosity of output                                                                                                                                               |
| suppressCPSWarnings    | Boolean     | Suppress duplicate warnings generated using ARM_CP_CPSI or ARM_CP_CPSD message identifiers                                                                                |
| showHiddenRegs         | Boolean     | Show hidden registers during register tracing                                                                                                                             |
| UAL                    | Boolean     | Disassemble using UAL syntax                                                                                                                                              |
| disableGICModel        | Boolean     | Disable the internal GIC model entirely                                                                                                                                   |
| enableGICv2.64kB.Page  | Boolean     | Enable 64kB page size for GICv2 memory-mapped register groups (Xilinx Zynq Ultra-scale support)                                                                           |
| enableVFPAAtReset      | Boolean     | Enable vector floating point (SIMD and VFP) instructions at reset. (Enables cp10/11 in CPACR and sets FPEXC.EN)                                                           |
| useInternalTCMs        | Boolean     | Enable internally-modeled TCM memories                                                                                                                                    |
| enableSystemBus        | Boolean     | Add 32-bit artifact System bus port, allowing system registers to be externally implemented                                                                               |
| enableSystemMonitorBus | Boolean     | Add 32-bit artifact SystemMonitor bus port, allowing system register accesses to be externally monitored                                                                  |
| compatibility          | Enumeration | Specify compatibility mode                                                                                                                                                |
|                        | ISA         |                                                                                                                                                                           |
|                        | gdb         |                                                                                                                                                                           |
|                        | nopSVC      |                                                                                                                                                                           |
| unpredictableR15       | Enumeration | Specify behavior for UNPREDICTABLE uses of AArch32 R15 register                                                                                                           |
|                        | undefined   |                                                                                                                                                                           |
|                        | nop         |                                                                                                                                                                           |
|                        | raz_wi      |                                                                                                                                                                           |
|                        | execute     |                                                                                                                                                                           |
|                        | assert      |                                                                                                                                                                           |
| unpredictableModal     | Enumeration | Specify behavior for UNPREDICTABLE instructions in certain AArch32 modes (for example, MRS using SPSR in System mode)                                                     |
|                        | undefined   |                                                                                                                                                                           |
|                        | nop         |                                                                                                                                                                           |
|                        | assert      |                                                                                                                                                                           |
| maxSIMDUnroll          | Uns32       | If SIMD operations are supported, specify the maximum number of parallel SIMD operations to unroll (unrolled operations can be faster, but produce more verbose JIT code) |

|                            |         |                                                                                                                                                                                      |
|----------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| override_debugMask         | Uns32   | Specifies debug mask, enabling debug output for model components                                                                                                                     |
| ASIDCacheSize              | Uns32   | Specifies the number of different ASIDs for which TLB entries are cached; a value of 0 implies no limit                                                                              |
| thumbNoCond                | Boolean | Specify whether trapped Thumb instructions set CV=1 and COND field in syndrome (if False, both are zero)                                                                             |
| enableHostAtomics          | Boolean | Enable use of host atomic instructions to emulate load/store exclusive operations (not architecturally accurate, accelerates parallel simulation)                                    |
| override_numCPUs           | Uns32   | Specify the number of cores in a multiprocessor (maximum of 8 for GICv1/GICv2)                                                                                                       |
| override_affinityMask      | Uns32   | Specify bitmask of implemented affinity bits in format Aff3:Aff2:Aff1:Aff0 (each a byte)                                                                                             |
| override_MPIDR_MT          | Boolean | Specifies that processor is multithreaded                                                                                                                                            |
| override_MPIDR_Aff0        | Uns32   | Override Aff0 field in MPIDR/MPIDR_EL1 register                                                                                                                                      |
| override_MPIDR_Aff1        | Uns32   | Override Aff1 field in MPIDR/MPIDR_EL1 register (also possible by writing CLUSTERIDAFF1 configuration net)                                                                           |
| override_MPIDR_Aff2        | Uns32   | Override Aff2 field in MPIDR/MPIDR_EL1 register (also possible by writing CLUSTERIDAFF2 configuration net)                                                                           |
| override_MPIDR_Aff3        | Uns32   | Override Aff3 field in MPIDR_EL1 register (also possible by writing CLUSTERIDAFF3 configuration net)                                                                                 |
| override_fcsePresent       | Boolean | Specifies that FCSE is present (if true)                                                                                                                                             |
| override_fpxcDexPresent    | Boolean | Specifies that the FPEXC.DEX register field is implemented (if true)                                                                                                                 |
| override_advSIMDPresent    | Boolean | Specifies that Advanced SIMD extensions are present (if true)                                                                                                                        |
| override_vfpPresent        | Boolean | Specifies that VFP extensions are present (if true)                                                                                                                                  |
| override_timerScaleFactor  | Uns32   | Specifies the fraction of MIPS rate to use for MPCore timers (generic timers or global/local/watchdogs depending on implementation). Defaults to 20 for generic timers, 2 for others |
| override_GICD_NSACRPresent | Boolean | Specifies that optional GICD_NSACR distributor registers are present (GICv2 only)                                                                                                    |
| override_GICD_PPISRPresent | Boolean | Specifies that implementation-specific GICD_PPISR distributor register is present (GICv1 ICDPPIS/ICPPISR, GICv1 and GICv2 only)                                                      |
| override_GICD_SPISRPresent | Boolean | Specifies that implementation-specific GICD_SPISR distributor registers are present (GICv1 ICDSPIS/ICSPISR)                                                                          |
| override_GIC_PPIMask       | Uns32   | Specify bitmask of implemented PPIs in the GIC (e.g. ID16 is 0x0001, ID31 is 0x8000)                                                                                                 |
| override_GICCDISABLE       | Boolean | Specify initial value of GICCDISABLE                                                                                                                                                 |
| override_SCTLR_V           | Boolean | Override SCTLR.V with the passed value (enables high vectors; also configurable using VINITHI pin)                                                                                   |
| override_SCTLR_IE          | Boolean | Override SCTLR.IE with the passed value (configures instruction endianness; also configurable using CFGIE pin)                                                                       |

|                                |         |                                                                                                                               |
|--------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------|
| override_SCTLR_EE              | Boolean | Override SCTLR.EE with the passed value (configures exception data endianness; also configurable using CFGEE pin)             |
| override_SCTLR_TE              | Boolean | Override SCTLR.TE with the passed value (configures Thumb state for exception handling; also configurable using TEINIT pin)   |
| override_SCTLR_NMFI            | Boolean | Override SCTLR.NMFI with the passed value (configures NMFI state for exception handling; also configurable using CFGNMFI pin) |
| override_SCTLR_CP15BEN_Present | Boolean | Enable ARMv7 SCTLR.CP15BEN bit (CP15 barrier enable)                                                                          |
| override_MIDR                  | Uns32   | Override MIDR/MIDR_EL1 register                                                                                               |
| override_CTR                   | Uns32   | Override CTR/CTR_EL0 register                                                                                                 |
| override_TLBTR                 | Uns32   | Override TLBTR register                                                                                                       |
| override_MPUIR                 | Uns32   | Override MPUIR register                                                                                                       |
| override_MPUIR_DRegion         | Uns32   | Override MPUIR.DRegion with the passed value                                                                                  |
| override_CLIDR                 | Uns32   | Override CLIDR/CLIDR_EL1 register                                                                                             |
| override_ACTLR_SMP             | Boolean | Override ACTLR SMP enable bit                                                                                                 |
| override_TCMTR                 | Uns32   | Override TCMTR register                                                                                                       |
| override_AIDR                  | Uns32   | Override AIDR/AIDR_EL1 register                                                                                               |
| override_ITCMRR_E              | Boolean | Override ITCMRR register enable bit                                                                                           |
| override_ITCMRR_SIZE           | Uns32   | Override ITCMRR register size bits                                                                                            |
| override_tcmsPresent           | Boolean | Specifies that TCM is present (if true)                                                                                       |
| override_DTCMRR_SIZE           | Uns32   | Override DTCMRR register size bits                                                                                            |
| override_CBAR                  | Uns32   | Override Configuration Base Address Register (corresponds to value on PERIPHBASE configuration input)                         |
| override_PFR0                  | Uns32   | Override ID_PFR0/ID_PFR0_EL1 register                                                                                         |
| override_PFR1                  | Uns32   | Override ID_PFR1/ID_PFR1_EL1 register                                                                                         |
| override_DFR0                  | Uns32   | Override ID_DFR0/ID_DFR0_EL1 register                                                                                         |
| override_AFR0                  | Uns32   | Override ID_AFR0/ID_AFR0_EL1 register                                                                                         |
| override_MMFR0                 | Uns32   | Override ID_MMFR0/ID_MMFR0_EL1 register                                                                                       |
| override_MMFR1                 | Uns32   | Override ID_MMFR1/ID_MMFR1_EL1 register                                                                                       |
| override_MMFR2                 | Uns32   | Override ID_MMFR2/ID_MMFR2_EL1 register                                                                                       |
| override_MMFR3                 | Uns32   | Override ID_MMFR3/ID_MMFR3_EL1 register                                                                                       |
| override_ISAR0                 | Uns32   | Override ID_ISAR0/ID_ISAR0_EL1 register                                                                                       |
| override_ISAR1                 | Uns32   | Override ID_ISAR1/ID_ISAR1_EL1 register                                                                                       |
| override_ISAR2                 | Uns32   | Override ID_ISAR2/ID_ISAR2_EL1 register                                                                                       |
| override_ISAR3                 | Uns32   | Override ID_ISAR3/ID_ISAR3_EL1 register                                                                                       |
| override_ISAR4                 | Uns32   | Override ID_ISAR4/ID_ISAR4_EL1 register                                                                                       |
| override_ISAR5                 | Uns32   | Override ID_ISAR5/ID_ISAR5_EL1 register                                                                                       |
| override_PMCR                  | Uns32   | Override PMCR/PMCR_EL0 register (not functionally significant in the model)                                                   |
| override_PMCEID0               | Uns64   | Override PMCEID0/PMCEID0_EL0 register (not functionally significant in the model)                                             |
| override_PMCEID1               | Uns64   | Override PMCEID1/PMCEID1_EL0 register (not functionally significant in the model)                                             |
| override_SACTLR                | Uns32   | Override Cortex-R profile SACTLR register (not functionally significant in the model)                                         |
| override_BuildOptions0         | Uns32   | Override Cortex-R profile BuildOptions0 register (not functionally significant in the model)                                  |

|                             |       |                                                                                              |
|-----------------------------|-------|----------------------------------------------------------------------------------------------|
| override_BuildOptions1      | Uns32 | Override Cortex-R profile BuildOptions1 register (not functionally significant in the model) |
| override_DBGDIDR            | Uns32 | Override DBGDIDR register (not functionally significant in the model)                        |
| override_FPSID              | Uns32 | Override SIMD/VFP FPSID register                                                             |
| override_MVFR0              | Uns32 | Override SIMD/VFP MVFR0/MVFR0_EL1 register                                                   |
| override_MVFR1              | Uns32 | Override SIMD/VFP MVFR1/MVFR1_EL1 register                                                   |
| override_FPEXC              | Uns32 | Override SIMD/VFP FPEXC/FPEXC32_EL2 register                                                 |
| override_GICC_IIDR          | Uns32 | Override GICC_IIDR register (GICv1 ICCI-IDR)                                                 |
| override_GICD_TYPER         | Uns32 | Override GICD_TYPER register (GICv1 IC-DICTR)                                                |
| override_GICD_TYPER_ITLines | Uns32 | Override ITLinesNumber field of GICD_TYPER register (GICv1 ICDICTR)                          |
| override_GICD_ICFGRN        | Uns32 | Override reset value of GICD_ICFGR2...GICD_ICFGRn (GICv1 ICDICFR2...ICDICFRn)                |
| override_GICD_IIDR          | Uns32 | Override GICD_IIDR register (GICv1 ICDI-IDR)                                                 |
| override_GICH_VTR           | Uns32 | Override GICH_VTR register                                                                   |
| override_ICCPMRBits         | Uns32 | Specify the number of writable bits in GICC_PMR (GICv1 ICCPMR)                               |
| override_minICCBPR          | Uns32 | Specify the minimum possible value for GICC_BPR (GICv1 ICCBPR)                               |
| override_ERG                | Uns32 | Specifies exclusive reservation granule                                                      |
| override_CCSIDR_1I          | Uns64 | Override CCSIDR/CCSIDR_EL1 (level 1 instruction)                                             |
| override_CCSIDR_1D          | Uns64 | Override CCSIDR/CCSIDR_EL1 (level 1 data)                                                    |
| override_CCSIDR_2I          | Uns64 | Override CCSIDR/CCSIDR_EL1 (level 2 instruction)                                             |
| override_CCSIDR_2D          | Uns64 | Override CCSIDR/CCSIDR_EL1 (level 2 data)                                                    |
| override_CCSIDR_3I          | Uns64 | Override CCSIDR/CCSIDR_EL1 (level 3 instruction)                                             |
| override_CCSIDR_3D          | Uns64 | Override CCSIDR/CCSIDR_EL1 (level 3 data)                                                    |
| override_CCSIDR_4I          | Uns64 | Override CCSIDR/CCSIDR_EL1 (level 4 instruction)                                             |
| override_CCSIDR_4D          | Uns64 | Override CCSIDR/CCSIDR_EL1 (level 4 data)                                                    |
| override_CCSIDR_5I          | Uns64 | Override CCSIDR/CCSIDR_EL1 (level 5 instruction)                                             |
| override_CCSIDR_5D          | Uns64 | Override CCSIDR/CCSIDR_EL1 (level 5 data)                                                    |
| override_CCSIDR_6I          | Uns64 | Override CCSIDR/CCSIDR_EL1 (level 6 instruction)                                             |
| override_CCSIDR_6D          | Uns64 | Override CCSIDR/CCSIDR_EL1 (level 6 data)                                                    |
| override_CCSIDR_7I          | Uns64 | Override CCSIDR/CCSIDR_EL1 (level 7 instruction)                                             |
| override_CCSIDR_7D          | Uns64 | Override CCSIDR/CCSIDR_EL1 (level 7 data)                                                    |

|                                              |         |                                                                                                                                          |
|----------------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------|
| override_STRoffsetPC12                       | Boolean | Specifies that STR/STR of PC should do so with 12:byte offset from the current instruction (if true), otherwise an 8:byte offset is used |
| override_mpuV5ExtAP                          | Boolean | Specifies that ARMv5 MPU extended access permissions implemented (cp15/5/2 and cp15/5/2)                                                 |
| override_fcseRequiresMMU                     | Boolean | Specifies that FCSE is active only when MMU is enabled (if true)                                                                         |
| override_ignoreBadCp15                       | Boolean | Specifies whether invalid coprocessor 15 access should be ignored (if true) or cause Invalid Instruction exceptions (if false)           |
| override_SGIDisable                          | Boolean | Override whether GIC SGIs may be disabled (if true) or are permanently enabled (if false)                                                |
| override_condUndefined                       | Boolean | Force undefined instructions to take Undefined Instruction exception even if they are conditional                                        |
| override_deviceStrongAligned                 | Boolean | Force accesses to Device and Strongly Ordered regions to be aligned                                                                      |
| override_Control_V                           | Boolean | Override SCTLR.V with the passed value (deprecated, use override_SCTLR.V)                                                                |
| override_MainId                              | Uns32   | Override MIDR register (deprecated, use override_MIDR)                                                                                   |
| override_CacheType                           | Uns32   | Override CTR register (deprecated, use override_CTR)                                                                                     |
| override_TLBType                             | Uns32   | Override TLBTR register (deprecated, use override_TLBTR)                                                                                 |
| override_MPUType                             | Uns32   | Override MPUIR register (deprecated, use override_MPUIR)                                                                                 |
| override_InstructionAttributes0              | Uns32   | Override ID_ISAR0 register (deprecated, use override_ISAR0)                                                                              |
| override_InstructionAttributes1              | Uns32   | Override ID_ISAR1 register (deprecated, use override_ISAR1)                                                                              |
| override_InstructionAttributes2              | Uns32   | Override ID_ISAR2 register (deprecated, use override_ISAR2)                                                                              |
| override_InstructionAttributes3              | Uns32   | Override ID_ISAR3 register (deprecated, use override_ISAR3)                                                                              |
| override_InstructionAttributes4              | Uns32   | Override ID_ISAR4 register (deprecated, use override_ISAR4)                                                                              |
| override_InstructionAttributes5              | Uns32   | Override ID_ISAR5 register (deprecated, use override_ISAR5)                                                                              |
| override_SCUControl_Address_filtering_enable | Boolean | Override SCU Control Address filtering enable bit in register                                                                            |
| override_MasterFilteringStart                | Uns32   | Override MasterFilteringStart register                                                                                                   |
| override_MasterFilteringEnd                  | Uns32   | Override MasterFilteringEnd register                                                                                                     |
| override_PeripheralsFilteringStart           | Uns32   | Override PeripheralsFilteringStart register                                                                                              |
| override_PeripheralsFilteringEnd             | Uns32   | Override PeripheralsFilteringEnd register                                                                                                |
| override_FPPFilteringStart0                  | Uns32   | Override FPP0 Port filtering start address                                                                                               |
| override_FPPFilteringEnd0                    | Uns32   | Override FPP0 Port filtering end address                                                                                                 |
| override_FPPFilteringStart1                  | Uns32   | Override FPP1 Port filtering start address                                                                                               |
| override_FPPFilteringEnd1                    | Uns32   | Override FPP1 Port filtering end address                                                                                                 |
| override_FPPFilteringStart2                  | Uns32   | Override FPP2 Port filtering start address                                                                                               |
| override_FPPFilteringEnd2                    | Uns32   | Override FPP2 Port filtering end address                                                                                                 |
| override_FPPFilteringStart3                  | Uns32   | Override FPP3 Port filtering start address                                                                                               |
| override_FPPFilteringEnd3                    | Uns32   | Override FPP3 Port filtering end address                                                                                                 |

Table 8.1: Parameters that can be set in: MPCORE

## 8.1 Parameter values and limits

These are the formal parameter limits and actual parameter values

| Name                       | Min | Max        | Default   | Actual        |
|----------------------------|-----|------------|-----------|---------------|
| <b>(Others)</b>            |     |            |           |               |
| variant                    |     |            |           | Cortex-R8MPx3 |
| verbose                    |     |            | t         | t             |
| suppressCPSWarnings        |     |            | f         | f             |
| showHiddenRegs             |     |            | f         | f             |
| UAL                        |     |            | t         | t             |
| disableGICModel            |     |            | f         | f             |
| enableGICv2_64kB_Page      |     |            | f         | f             |
| enableVFPAAtReset          |     |            | f         | f             |
| useInternalTCMs            |     |            | f         | f             |
| enableSystemBus            |     |            | f         | f             |
| enableSystemMonitorBus     |     |            | f         | f             |
| compatibility              |     |            | ISA       | ISA           |
| unpredictableR15           |     |            | undefined | undefined     |
| unpredictableModal         |     |            | nop       | nop           |
| maxSIMDUnroll              | 1   | 16         | 2         | 2             |
| override_debugMask         | 0   | 0xffffffff | 0         | 0             |
| ASIDCacheSize              | 0   | 0x100      | 8         | 8             |
| thumbNoCond                |     |            | f         | f             |
| enableHostAtomics          |     |            | f         | f             |
| endian                     |     |            | none      | none          |
| override_numCPUs           | 0   | 32         | 3         | 3             |
| override_affinityMask      | 0   | 0xffffffff | 0         | 0             |
| override_MPIDR_MT          |     |            | f         | f             |
| override_MPIDR_Aff0        | 0   | 255        | 0         | 0             |
| override_MPIDR_Aff1        | 0   | 255        | 0         | 0             |
| override_MPIDR_Aff2        | 0   | 255        | 0         | 0             |
| override_MPIDR_Aff3        | 0   | 255        | 0         | 0             |
| override_fcsePresent       |     |            | t         | t             |
| override_fpexcDexPresent   |     |            | f         | f             |
| override_advSIMDPresent    |     |            | f         | f             |
| override_vfpPresent        |     |            | f         | f             |
| override_timerScaleFactor  | 1   | 0x1ff      | 2         | 2             |
| override_GICD_NSACRPresent |     |            | f         | f             |
| override_GICD_PPISRPresent |     |            | f         | f             |
| override_GICD_SPISRPresent |     |            | f         | f             |
| override_GIC_PPIMask       | 0   | 0xffff     | 0         | 0             |
| override_GICCDISABLE       |     |            | f         | f             |
| override_SCTLR_V           |     |            | f         | f             |
| override_SCTLR_IE          |     |            | f         | f             |
| override_SCTLR_EE          |     |            | f         | f             |

|                                |   |                  |            |            |
|--------------------------------|---|------------------|------------|------------|
| override_SCTLR_TE              |   |                  | f          | f          |
| override_SCTLR_NMFI            |   |                  | f          | f          |
| override_SCTLR_CP15BEN_Present |   |                  | f          | f          |
| override_MIDR                  | 0 | 0xffffffff       | 0x410fc183 | 0x410fc183 |
| override_CTR                   | 0 | 0xffffffff       | 0x8333c003 | 0x8333c003 |
| override_TLBTR                 | 0 | 0xffffffff       | 0          | 0          |
| override_MPUIR                 | 0 | 0xffffffff       | 0xc00      | 0xc00      |
| override_MPUIR_DRegion         | 0 | 0xffffffff       | 12         | 12         |
| override_CLIDR                 | 0 | 0xffffffff       | 0x9200003  | 0x9200003  |
| override_ACTLR_SMP             |   |                  | f          | f          |
| override_TCMTR                 | 0 | 0xffffffff       | 0x10001    | 0x10001    |
| override_AIDR                  | 0 | 0xffffffff       | 0          | 0          |
| override_ITCMRR_E              |   |                  | f          | f          |
| override_ITCMRR_SIZE           | 0 | 0xffffffff       | 4          | 4          |
| override_tcmsPresent           |   |                  | t          | t          |
| override_DTCMRR_SIZE           | 0 | 0xffffffff       | 4          | 4          |
| override_CBAR                  | 0 | 0xffffffff       | 0xae000000 | 0xae000000 |
| override_PFR0                  | 0 | 0xffffffff       | 0x131      | 0x131      |
| override_PFR1                  | 0 | 0xffffffff       | 1          | 1          |
| override_DFR0                  | 0 | 0xffffffff       | 0x10404    | 0x10404    |
| override_AFR0                  | 0 | 0xffffffff       | 0          | 0          |
| override_MMFR0                 | 0 | 0xffffffff       | 0x110130   | 0x110130   |
| override_MMFR1                 | 0 | 0xffffffff       | 0          | 0          |
| override_MMFR2                 | 0 | 0xffffffff       | 0x1200000  | 0x1200000  |
| override_MMFR3                 | 0 | 0xffffffff       | 0x2111     | 0x2111     |
| override_ISAR0                 | 0 | 0xffffffff       | 0x2101111  | 0x2101111  |
| override_ISAR1                 | 0 | 0xffffffff       | 0x13112111 | 0x13112111 |
| override_ISAR2                 | 0 | 0xffffffff       | 0x21232141 | 0x21232141 |
| override_ISAR3                 | 0 | 0xffffffff       | 0x1112131  | 0x1112131  |
| override_ISAR4                 | 0 | 0xffffffff       | 0x10142    | 0x10142    |
| override_ISAR5                 | 0 | 0xffffffff       | 0          | 0          |
| override_PMCR                  | 0 | 0xffffffff       | 0x41182000 | 0x41182000 |
| override_PMCEID0               | 0 | 0xffffffffffffff | 0          | 0          |
| override_PMCEID1               | 0 | 0xffffffffffffff | 0          | 0          |
| override_SACTLR                | 0 | 0xffffffff       | 0          | 0          |
| override_BuildOptions0         | 0 | 0xffffffff       | 0          | 0          |
| override_BuildOptions1         | 0 | 0xffffffff       | 0          | 0          |
| override_DBGDIDR               | 0 | 0xffffffff       | 0          | 0          |
| override_FPSID                 | 0 | 0xffffffff       | 0x41023180 | 0x41023180 |
| override_MVFR0                 | 0 | 0xffffffff       | 0x10110221 | 0x10110221 |
| override_MVFR1                 | 0 | 0xffffffff       | 0x1000011  | 0x1000011  |
| override_FPEXC                 | 0 | 0xffffffff       | 0          | 0          |
| override_GICC_IIDR             | 0 | 0xffffffff       | 0x3901043b | 0x3901043b |
| override_GICD_TYPER            | 0 | 0xffffffff       | 2          | 2          |
| override_GICD_TYPER_ITLines    | 0 | 31               | 2          | 2          |

|                                              |   |                  |            |            |
|----------------------------------------------|---|------------------|------------|------------|
| override_GICD_ICFGRN                         | 0 | 0xffffffff       | 0          | 0          |
| override_GICD_IIDR                           | 0 | 0xffffffff       | 0x400043b  | 0x400043b  |
| override_GICH_VTR                            | 0 | 0xffffffff       | 0          | 0          |
| override_ICCPMRBits                          | 4 | 8                | 5          | 5          |
| override_minICCBPR                           | 0 | 7                | 2          | 2          |
| override_ERG                                 | 3 | 11               | 3          | 3          |
| override_CCSIDR_1I                           | 0 | 0xffffffffffffff | 0x703fe019 | 0x703fe019 |
| override_CCSIDR_1D                           | 0 | 0xffffffffffffff | 0x703fe019 | 0x703fe019 |
| override_CCSIDR_2I                           | 0 | 0xffffffffffffff | 0          | 0          |
| override_CCSIDR_2D                           | 0 | 0xffffffffffffff | 0          | 0          |
| override_CCSIDR_3I                           | 0 | 0xffffffffffffff | 0          | 0          |
| override_CCSIDR_3D                           | 0 | 0xffffffffffffff | 0          | 0          |
| override_CCSIDR_4I                           | 0 | 0xffffffffffffff | 0          | 0          |
| override_CCSIDR_4D                           | 0 | 0xffffffffffffff | 0          | 0          |
| override_CCSIDR_5I                           | 0 | 0xffffffffffffff | 0          | 0          |
| override_CCSIDR_5D                           | 0 | 0xffffffffffffff | 0          | 0          |
| override_CCSIDR_6I                           | 0 | 0xffffffffffffff | 0          | 0          |
| override_CCSIDR_6D                           | 0 | 0xffffffffffffff | 0          | 0          |
| override_CCSIDR_7I                           | 0 | 0xffffffffffffff | 0          | 0          |
| override_CCSIDR_7D                           | 0 | 0xffffffffffffff | 0          | 0          |
| override_STROffsetPC12                       |   |                  | t          | t          |
| override_mpuV5ExtAP                          |   |                  | f          | f          |
| override_fcseRequiresMMU                     |   |                  | f          | f          |
| override_ignoreBadCp15                       |   |                  | f          | f          |
| override_SGIDisable                          |   |                  | f          | f          |
| override_condUndefined                       |   |                  | f          | f          |
| override_deviceStrongAligned                 |   |                  | t          | t          |
| override_Control_V                           |   |                  | f          | f          |
| override_MainId                              | 0 | 0xffffffff       | 0x410fc183 | 0x410fc183 |
| override_CacheType                           | 0 | 0xffffffff       | 0x8333c003 | 0x8333c003 |
| override_TLBType                             | 0 | 0xffffffff       | 0          | 0          |
| override MPUType                             | 0 | 0xffffffff       | 0xc00      | 0xc00      |
| override_InstructionAttributes0              | 0 | 0xffffffff       | 0x2101111  | 0x2101111  |
| override_InstructionAttributes1              | 0 | 0xffffffff       | 0x13112111 | 0x13112111 |
| override_InstructionAttributes2              | 0 | 0xffffffff       | 0x21232141 | 0x21232141 |
| override_InstructionAttributes3              | 0 | 0xffffffff       | 0x1112131  | 0x1112131  |
| override_InstructionAttributes4              | 0 | 0xffffffff       | 0x10142    | 0x10142    |
| override_InstructionAttributes5              | 0 | 0xffffffff       | 0          | 0          |
| override_SCUControl_Address_filtering_enable |   |                  | f          | f          |
| override_MasterFilteringStart                | 0 | 0xffffffff       | 0xff00000  | 0xff00000  |
| override_MasterFilteringEnd                  | 0 | 0xffffffff       | 0          | 0          |
| override_PeripheralsFilteringStart           | 0 | 0xffffffff       | 0xff00000  | 0xff00000  |
| override_PeripheralsFilteringEnd             | 0 | 0xffffffff       | 0          | 0          |
| override_FPPFilteringStart0                  | 0 | 0xffffffff       | 0xff00000  | 0xff00000  |
| override_FPPFilteringEnd0                    | 0 | 0xffffffff       | 0          | 0          |



|                             |   |            |            |            |
|-----------------------------|---|------------|------------|------------|
| override_FPPFilteringStart1 | 0 | 0xffffffff | 0xfff00000 | 0xfff00000 |
| override_FPPFilteringEnd1   | 0 | 0xffffffff | 0          | 0          |
| override_FPPFilteringStart2 | 0 | 0xffffffff | 0xfff00000 | 0xfff00000 |
| override_FPPFilteringEnd2   | 0 | 0xffffffff | 0          | 0          |
| override_FPPFilteringStart3 | 0 | 0xffffffff | 0xfff00000 | 0xfff00000 |
| override_FPPFilteringEnd3   | 0 | 0xffffffff | 0          | 0          |

Table 8.2: Parameter values and limits

## Chapter 9

# Execution Modes

| Mode       | Code |
|------------|------|
| User       | 16   |
| FIQ        | 17   |
| IRQ        | 18   |
| Supervisor | 19   |
| Abort      | 23   |
| Undefined  | 27   |
| System     | 31   |

Table 9.1: Modes implemented in: CPU

## Chapter 10

# Exceptions

| <b>Exception</b> | Code |
|------------------|------|
| Reset            | 0    |
| Undefined        | 1    |
| SupervisorCall   | 2    |
| PrefetchAbort    | 5    |
| DataAbort        | 6    |
| IRQ              | 8    |
| FIQ              | 9    |

Table 10.1: Exceptions implemented in: CPU

# Chapter 11

## Hierarchy of the model

A CPU core may be configured to instance many processors of a Symmetrical Multi Processor (SMP). A CPU core may also have sub elements within a processor, for example hardware threading blocks.

OVP processor models can be written to include SMP blocks and to have many levels of hierarchy. Some OVP CPU models may have a fixed hierarchy, and some may be configured by settings in a configuration register. Please see the register definitions of this model.

This model documentation shows the settings and hierarchy of the default settings for this model variant.

### 11.1 Level 1: MPCORE

This level in the model hierarchy has 2 commands.

This level in the model hierarchy has no register groups.

This level in the model hierarchy has 3 children:

CPU0, CPU1 and CPU2.

### 11.2 Level 2: CPU

This level in the model hierarchy has 4 commands.

This level in the model hierarchy has 17 register groups:

| Group name         | Registers |
|--------------------|-----------|
| Core               | 16        |
| Control            | 3         |
| User               | 7         |
| FIQ                | 8         |
| IRQ                | 3         |
| Supervisor         | 3         |
| Undefined          | 3         |
| Abort              | 3         |
| SIMD_VFP           | 16        |
| SIMD_VFP_SYS       | 5         |
| Coprocessor_32_bit | 80        |

|                            |    |
|----------------------------|----|
| Integration_support        | 5  |
| MPCore_SCR                 | 18 |
| MPCore_distributor         | 85 |
| MPCore_processor_interface | 8  |
| MPCore_timer_and_watchdog  | 10 |
| MPCore_global_timer        | 5  |

Table 11.1: Register groups

This level in the model hierarchy has no children.

# Chapter 12

## Model Commands

A Processor model can implement one or more **Model Commands** available to be invoked from the simulator command line, from the OP API or from the Imperas Multiprocessor Debugger.

### 12.1 Level 1: MPCORE

#### 12.1.1 isync

specify instruction address range for synchronous execution

| Argument   | Type  | Description                                  |
|------------|-------|----------------------------------------------|
| -addresshi | Uns64 | end address of synchronous execution range   |
| -addresslo | Uns64 | start address of synchronous execution range |

Table 12.1: isync command arguments

#### 12.1.2 itrace

enable or disable instruction tracing

| Argument          | Type    | Description                                                                                                                                           |
|-------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| -access           | String  | show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system)                     |
| -after            | Uns64   | apply after this many instructions                                                                                                                    |
| -enable           | Boolean | enable instruction tracing                                                                                                                            |
| -full             | Boolean | turn on all trace features                                                                                                                            |
| -instructioncount | Boolean | include the instruction number in each trace                                                                                                          |
| -memory           | String  | (Alias for access). show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system) |
| -mode             | Boolean | show processor mode changes                                                                                                                           |
| -off              | Boolean | disable instruction tracing                                                                                                                           |
| -on               | Boolean | enable instruction tracing                                                                                                                            |
| -processorname    | Boolean | Include processor name in all trace lines                                                                                                             |

|                 |         |                                            |
|-----------------|---------|--------------------------------------------|
| -registerchange | Boolean | show registers changed by this instruction |
| -registers      | Boolean | show registers after each trace            |

Table 12.2: itrace command arguments

## 12.2 Level 2: CPU

### 12.2.1 debugflags

show or modify the processor debug flags

| Argument | Type    | Description                                                 |
|----------|---------|-------------------------------------------------------------|
| -get     | Boolean | print current processor flags value                         |
| -mask    | Boolean | print valid debug flag bits                                 |
| -set     | Int32   | new processor flags (only flags 0x000003e4 can be modified) |

Table 12.3: debugflags command arguments

### 12.2.2 isync

specify instruction address range for synchronous execution

| Argument   | Type  | Description                                  |
|------------|-------|----------------------------------------------|
| -addresshi | Uns64 | end address of synchronous execution range   |
| -addresslo | Uns64 | start address of synchronous execution range |

Table 12.4: isync command arguments

### 12.2.3 itrace

enable or disable instruction tracing

| Argument          | Type    | Description                                                                                                                                           |
|-------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| -access           | String  | show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system)                     |
| -after            | Uns64   | apply after this many instructions                                                                                                                    |
| -enable           | Boolean | enable instruction tracing                                                                                                                            |
| -full             | Boolean | turn on all trace features                                                                                                                            |
| -instructioncount | Boolean | include the instruction number in each trace                                                                                                          |
| -memory           | String  | (Alias for access). show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system) |
| -mode             | Boolean | show processor mode changes                                                                                                                           |
| -off              | Boolean | disable instruction tracing                                                                                                                           |
| -on               | Boolean | enable instruction tracing                                                                                                                            |

|                 |         |                                            |
|-----------------|---------|--------------------------------------------|
| -processorname  | Boolean | Include processor name in all trace lines  |
| -registerchange | Boolean | show registers changed by this instruction |
| -registers      | Boolean | show registers after each trace            |

Table 12.5: itrace command arguments

## 12.2.4 listSysRegsAA32

### 12.2.4.1 Argument description

List all AArch32 system registers



# Chapter 13

## Registers

### 13.1 Level 1: MPCORE

No registers.

### 13.2 Level 2: CPU

#### 13.2.1 Core

Registers at level:2, type:CPU group:Core

| Name | Bits | Initial-Hex | RW | Description     |
|------|------|-------------|----|-----------------|
| r0   | 32   | 0           | rw |                 |
| r1   | 32   | 0           | rw |                 |
| r2   | 32   | 0           | rw |                 |
| r3   | 32   | 0           | rw |                 |
| r4   | 32   | 0           | rw |                 |
| r5   | 32   | 0           | rw |                 |
| r6   | 32   | 0           | rw |                 |
| r7   | 32   | 0           | rw |                 |
| r8   | 32   | 0           | rw |                 |
| r9   | 32   | 0           | rw |                 |
| r10  | 32   | 0           | rw |                 |
| r11  | 32   | 0           | rw | frame pointer   |
| r12  | 32   | 0           | rw |                 |
| sp   | 32   | 0           | rw | stack pointer   |
| lr   | 32   | 0           | rw |                 |
| pc   | 32   | 0           | rw | program counter |

Table 13.1: Registers at level 2, type:CPU group:Core

#### 13.2.2 Control

Registers at level:2, type:CPU group:Control

| Name | Bits | Initial-Hex | RW | Description                  |
|------|------|-------------|----|------------------------------|
| fps  | 32   | 0           | rw | archaic FPSCR view (for gdb) |
| cpsr | 32   | 1d3         | rw |                              |
| spsr | 32   | 0           | rw |                              |

Table 13.2: Registers at level 2, type:CPU group:Control

### 13.2.3 User

Registers at level:2, type:CPU group:User

| Name    | Bits | Initial-Hex | RW | Description |
|---------|------|-------------|----|-------------|
| r8_usr  | 32   | 0           | rw |             |
| r9_usr  | 32   | 0           | rw |             |
| r10_usr | 32   | 0           | rw |             |
| r11_usr | 32   | 0           | rw |             |
| r12_usr | 32   | 0           | rw |             |
| sp_usr  | 32   | 0           | rw |             |
| lr_usr  | 32   | 0           | rw |             |

Table 13.3: Registers at level 2, type:CPU group:User

### 13.2.4 FIQ

Registers at level:2, type:CPU group:FIQ

| Name     | Bits | Initial-Hex | RW | Description |
|----------|------|-------------|----|-------------|
| r8_fiq   | 32   | 0           | rw |             |
| r9_fiq   | 32   | 0           | rw |             |
| r10_fiq  | 32   | 0           | rw |             |
| r11_fiq  | 32   | 0           | rw |             |
| r12_fiq  | 32   | 0           | rw |             |
| sp_fiq   | 32   | 0           | rw |             |
| lr_fiq   | 32   | 0           | rw |             |
| spsr_fiq | 32   | 0           | rw |             |

Table 13.4: Registers at level 2, type:CPU group:FIQ

### 13.2.5 IRQ

Registers at level:2, type:CPU group:IRQ

| Name     | Bits | Initial-Hex | RW | Description |
|----------|------|-------------|----|-------------|
| sp_irq   | 32   | 0           | rw |             |
| lr_irq   | 32   | 0           | rw |             |
| spsr_irq | 32   | 0           | rw |             |

Table 13.5: Registers at level 2, type:CPU group:IRQ

### 13.2.6 Supervisor

Registers at level:2, type:CPU group:Supervisor

| Name     | Bits | Initial-Hex | RW | Description |
|----------|------|-------------|----|-------------|
| sp_svc   | 32   | 0           | rw |             |
| lr_svc   | 32   | 0           | rw |             |
| spsr_svc | 32   | 0           | rw |             |

Table 13.6: Registers at level 2, type:CPU group:Supervisor

### 13.2.7 Undefined

Registers at level:2, type:CPU group:Undefined

| Name       | Bits | Initial-Hex | RW | Description |
|------------|------|-------------|----|-------------|
| sp_undef   | 32   | 0           | rw |             |
| lr_undef   | 32   | 0           | rw |             |
| spsr_undef | 32   | 0           | rw |             |

Table 13.7: Registers at level 2, type:CPU group:Undefined

### 13.2.8 Abort

Registers at level:2, type:CPU group:Abort

| Name     | Bits | Initial-Hex | RW | Description |
|----------|------|-------------|----|-------------|
| sp_abt   | 32   | 0           | rw |             |
| lr_abt   | 32   | 0           | rw |             |
| spsr_abt | 32   | 0           | rw |             |

Table 13.8: Registers at level 2, type:CPU group:Abort

### 13.2.9 SIMD\_VFP

Registers at level:2, type:CPU group:SIMD\_VFP

| Name | Bits | Initial-Hex | RW | Description |
|------|------|-------------|----|-------------|
| d0   | 64   | 0           | rw |             |
| d1   | 64   | 0           | rw |             |
| d2   | 64   | 0           | rw |             |
| d3   | 64   | 0           | rw |             |
| d4   | 64   | 0           | rw |             |
| d5   | 64   | 0           | rw |             |
| d6   | 64   | 0           | rw |             |
| d7   | 64   | 0           | rw |             |
| d8   | 64   | 0           | rw |             |
| d9   | 64   | 0           | rw |             |
| d10  | 64   | 0           | rw |             |
| d11  | 64   | 0           | rw |             |
| d12  | 64   | 0           | rw |             |
| d13  | 64   | 0           | rw |             |
| d14  | 64   | 0           | rw |             |
| d15  | 64   | 0           | rw |             |

Table 13.9: Registers at level 2, type:CPU group:SIMD\_VFP

### 13.2.10 SIMD\_VFP\_SYS

Registers at level:2, type:CPU group:SIMD\_VFP\_SYS

| Name  | Bits | Initial-Hex | RW | Description                   |
|-------|------|-------------|----|-------------------------------|
| FPSID | 32   | 41023180    | r- | floating-point system ID      |
| FPSCR | 32   | 0           | rw | floating-point status/control |
| FPEXC | 32   | 0           | rw | floating-point exception      |
| MVFR0 | 32   | 10110221    | r- | Media/VFP feature 0           |

|       |    |         |    |                     |
|-------|----|---------|----|---------------------|
| MVFR1 | 32 | 1000011 | r- | Media/VFP feature 1 |
|-------|----|---------|----|---------------------|

Table 13.10: Registers at level 2, type:CPU group:SIMD\_VFP\_SYS

### 13.2.11 Coprocessor\_32\_bit

Registers at level:2, type:CPU group:Coprocessor\_32\_bit

| Name       | Bits | Initial-Hex | RW | Description                                       |
|------------|------|-------------|----|---------------------------------------------------|
| ACTLR      | 32   | 0           | rw | Auxiliary Control                                 |
| ADFSR      | 32   | 0           | rw | Auxiliary Data Fault Status                       |
| AIFSR      | 32   | 0           | rw | Auxiliary Instruction Fault Status                |
| BPIALL     | 32   | -           | -w | Branch Predictor Invalidate All                   |
| BPIALLIS   | 32   | -           | -w | Branch Predictor Invalidate All (IS)              |
| BPIMVA     | 32   | -           | -w | Branch Predictor Invalidate by VA                 |
| CBAR       | 32   | ae000000    | rw | Configuration Base Address                        |
| CCSIDR     | 32   | 703fe019    | r- | Cache Size ID                                     |
| CLIDR      | 32   | 9200003     | r- | Cache Level ID                                    |
| CONTEXTIDR | 32   | 0           | rw | Context ID                                        |
| CP15DMB    | 32   | -           | -w | CP15 Data Memory Barrier                          |
| CP15DSB    | 32   | -           | -w | CP15 Data Synchronization Barrier                 |
| CP15ISB    | 32   | -           | -w | CP15 Instruction Synchronization Barrier          |
| CP15NOP    | 32   | -           | -w | CP15 NOP                                          |
| CPACR      | 32   | c0000000    | rw | Coprocessor Access Control                        |
| CSSELR     | 32   | 0           | rw | Cache Size Selection                              |
| CTR        | 32   | 8333c003    | r- | Cache Type                                        |
| DBGDIDR    | 32   | 0           | r- | Debug ID                                          |
| DCCIMVAC   | 32   | -           | -w | Data Cache Line Clean and Invalidate by VA to PoC |
| DCCISW     | 32   | -           | -w | Data Cache Line Clean and Invalidate by Set/Way   |
| DCCMVAC    | 32   | -           | -w | Data Cache Line Clean by VA to PoC                |
| DCCMVAU    | 32   | -           | -w | Data Cache Line Clean by VA to PoU                |
| DCCSW      | 32   | -           | -w | Data Cache Line Clean by Set/Way                  |
| DCIMVAC    | 32   | -           | -w | Data Cache Line Invalidate by VA to PoC           |
| DCISW      | 32   | -           | -w | Data Cache Line Invalidate by Set/Way             |
| DFAR       | 32   | 0           | rw | Data Fault Address                                |
| DFSR       | 32   | 0           | rw | Data Fault Status                                 |
| DRACR      | 32   | 0           | rw | Data Region Access Control                        |
| DRBAR      | 32   | 0           | rw | Data Region Base Address                          |
| DRSR       | 32   | 0           | rw | Data Region Size/Enable                           |
| DTCMRR     | 32   | 10          | rw | DTCM Region                                       |
| FCSEIDR    | 32   | 0           | rw | FCSE Process ID                                   |
| ICIALLU    | 32   | -           | -w | Instruction Cache Invalidate All                  |
| ICIALLUIS  | 32   | -           | -w | Instruction Cache Invalidate All (IS)             |
| ICIMVAU    | 32   | -           | -w | Instruction Cache Invalidate by VA                |
| ID.AFR0    | 32   | 0           | r- | Auxiliary Feature 0                               |
| ID.DFR0    | 32   | 10404       | r- | Debug Feature 0                                   |
| ID.ISAR0   | 32   | 2101111     | r- | Instruction Set Attribute 0                       |
| ID.ISAR1   | 32   | 13112111    | r- | Instruction Set Attribute 1                       |
| ID.ISAR2   | 32   | 21232141    | r- | Instruction Set Attribute 2                       |
| ID.ISAR3   | 32   | 1112131     | r- | Instruction Set Attribute 3                       |
| ID.ISAR4   | 32   | 10142       | r- | Instruction Set Attribute 4                       |
| ID.ISAR5   | 32   | 0           | r- | Instruction Set Attribute 5                       |
| ID.MMFR0   | 32   | 110130      | r- | Memory Model Feature 0                            |
| ID.MMFR1   | 32   | 0           | r- | Memory Model Feature 1                            |
| ID.MMFR2   | 32   | 1200000     | r- | Memory Model Feature 2                            |

|             |    |          |    |                                              |
|-------------|----|----------|----|----------------------------------------------|
| ID_MMFR3    | 32 | 2111     | r- | Memory Model Feature 3                       |
| ID_PFR0     | 32 | 131      | r- | Processor Feature 0                          |
| ID_PFR1     | 32 | 1        | r- | Processor Feature 1                          |
| IFAR        | 32 | 0        | rw | Instruction Fault Address                    |
| IFSR        | 32 | 0        | rw | Instruction Fault Status                     |
| ISR         | 32 | 0        | r- | Interrupt Status                             |
| ITCMRR      | 32 | 10       | rw | ITCM Region                                  |
| JIDR        | 32 | 0        | rw | Jazelle ID                                   |
| JMCR        | 32 | 0        | rw | Jazelle Main Configuration                   |
| JOSCR       | 32 | 0        | rw | Jazelle OS Control                           |
| L2CTLR      | 32 | 2000000  | rw | L2 Control                                   |
| L2ECTLR     | 32 | 0        | rw | L2 Extended Control                          |
| MIDR        | 32 | 410fc183 | r- | Main ID                                      |
| MPIDR       | 32 | 80000000 | r- | Multiprocessor Affinity                      |
| MPUIR       | 32 | c00      | r- | MPU Type                                     |
| PAR         | 32 | 0        | rw | Physical Address                             |
| PMCCNTR     | 32 | 0        | rw | Performance Monitors Cycle Count             |
| PMCNTENCLR  | 32 | 0        | rw | Performance Monitors Count Enable Clear      |
| PMCNTENSET  | 32 | 0        | rw | Performance Monitors Count Enable Set        |
| PMCR        | 32 | 41182000 | rw | Performance Monitors Control                 |
| PMINTENCLR  | 32 | 0        | rw | Performance Monitors Interrupt Enable Clear  |
| PMINTENSET  | 32 | 0        | rw | Performance Monitors Interrupt Enable Set    |
| PMOVSr      | 32 | 0        | rw | Performance Monitors Overflow Flag Status    |
| PMSELR      | 32 | 0        | rw | Performance Monitors Event Counter Selection |
| PMSWINC     | 32 | -        | -w | Performance Monitors Software Increment      |
| PMUSERENR   | 32 | 0        | rw | Performance Monitors User Enable             |
| PMXEVCNTR   | 32 | 0        | rw | Performance Monitors Selected Event Count    |
| PMXEVTPYPER | 32 | 0        | rw | Performance Monitors Selected Event Type     |
| RGNR        | 32 | 0        | rw | Region Number                                |
| SCTLR       | 32 | 0        | rw | System Control                               |
| TCMTR       | 32 | 10001    | r- | TCM Type                                     |
| TPIDRPRW    | 32 | 0        | rw | PL0 Read/Write Software Thread ID            |
| TPIDRURO    | 32 | 0        | rw | PL0 Read-Only Software Thread ID             |
| TPIDRURW    | 32 | 0        | rw | PL1 Software Thread ID                       |

Table 13.11: Registers at level 2, type:CPU group:Coprocessor\_32\_bit

### 13.2.12 Integration\_support

Registers at level:2, type:CPU group:Integration\_support

| Name       | Bits | Initial-Hex | RW | Description                                             |
|------------|------|-------------|----|---------------------------------------------------------|
| transactPL | 32   | 1           | r- | privilege level of current memory transaction           |
| transactAT | 32   | 0           | r- | current memory transaction type: PA=1, VA=0             |
| HaltReason | 8    | 0           | r- | bit field indicating halt reason                        |
| atomicType | 8    | 0           | r- | current atomic instruction type (1:atomic, 2:exclusive) |
| WakeWFINow | 8    | 0           | -w | wake up core from WFI state                             |

Table 13.12: Registers at level 2, type:CPU group:Integration\_support

### 13.2.13 MPCore\_SCR

Registers at level:2, type:CPU group:MPCore\_SCR

| Name | Bits | Initial-Hex | RW | Description |
|------|------|-------------|----|-------------|
|------|------|-------------|----|-------------|

|                           |    |          |    |                               |
|---------------------------|----|----------|----|-------------------------------|
| FPPFilteringEnd0          | 32 | 0        | rw | FPP0 Filtering End Address    |
| FPPFilteringEnd1          | 32 | 0        | rw | FPP1 Filtering End Address    |
| FPPFilteringEnd2          | 32 | 0        | rw | FPP2 Filtering End Address    |
| FPPFilteringEnd3          | 32 | 0        | rw | FPP3 Filtering End Address    |
| FPPFilteringStart0        | 32 | fff00000 | rw | FPP0 Filtering Start Address  |
| FPPFilteringStart1        | 32 | fff00000 | rw | FPP1 Filtering Start Address  |
| FPPFilteringStart2        | 32 | fff00000 | rw | FPP2 Filtering Start Address  |
| FPPFilteringStart3        | 32 | fff00000 | rw | FPP3 Filtering Start Address  |
| MasterFilteringEnd        | 32 | 0        | rw | Master Filtering End          |
| MasterFilteringStart      | 32 | fff00000 | rw | Master Filtering Start        |
| PeripheralsFilteringEnd   | 32 | 0        | rw | Peripheral Filtering End      |
| PeripheralsFilteringStart | 32 | fff00000 | rw | Peripheral Filtering Start    |
| SCUCPUPowerStatus         | 32 | 3000000  | rw | SCU CPU Power Status          |
| SCUConfiguration          | 32 | fff02    | r- | SCU Configuration             |
| SCUControl                | 32 | 0        | rw | SCU Control                   |
| SCUInvalidateSecure       | 32 | 0        | -w | SCU Invalidate Secure         |
| SCUNSAC                   | 32 | 0        | rw | SCU Non-secure Access Control |
| SCUSAC                    | 32 | f        | rw | SCU Secure Access Control     |

Table 13.13: Registers at level 2, type:CPU group:MPCore.SCR

### 13.2.14 MPCore\_distributor

Registers at level:2, type:CPU group:MPCore\_distributor

| Name     | Bits | Initial-Hex | RW | Description                |
|----------|------|-------------|----|----------------------------|
| ICCIDR0  | 32   | d           | r- | Component ID 0             |
| ICCIDR1  | 32   | 90          | r- | Component ID 1             |
| ICCIDR2  | 32   | 5           | r- | Component ID 2             |
| ICCIDR3  | 32   | b1          | r- | Component ID 3             |
| ICDABR0  | 32   | 0           | rw | Interrupt Set-Active 0     |
| ICDABR1  | 32   | 0           | rw | Interrupt Set-Active 1     |
| ICDABR2  | 32   | 0           | rw | Interrupt Set-Active 2     |
| ICDDCR   | 32   | 0           | rw | Distributor Control        |
| ICDICER0 | 32   | fff         | rw | Interrupt Clear-Enable 0   |
| ICDICER1 | 32   | 0           | rw | Interrupt Clear-Enable 1   |
| ICDICER2 | 32   | 0           | rw | Interrupt Clear-Enable 2   |
| ICDICFR0 | 32   | aaaaaaaa    | rw | Interrupt Configuration 0  |
| ICDICFR1 | 32   | 7dc00000    | rw | Interrupt Configuration 1  |
| ICDICFR2 | 32   | 0           | rw | Interrupt Configuration 2  |
| ICDICFR3 | 32   | 0           | rw | Interrupt Configuration 3  |
| ICDICFR4 | 32   | 0           | rw | Interrupt Configuration 4  |
| ICDICFR5 | 32   | 0           | rw | Interrupt Configuration 5  |
| ICDICPR0 | 32   | 0           | rw | Interrupt Clear-Pending 0  |
| ICDICPR1 | 32   | 0           | rw | Interrupt Clear-Pending 1  |
| ICDICPR2 | 32   | 0           | rw | Interrupt Clear-Pending 2  |
| ICDICTR  | 32   | 42          | r- | Interrupt Controller Type  |
| ICDIIDR  | 32   | 400043b     | r- | Distributor Implementor ID |
| ICDIPR0  | 32   | 0           | rw | Interrupt Priority 0       |
| ICDIPR1  | 32   | 0           | rw | Interrupt Priority 1       |
| ICDIPR2  | 32   | 0           | rw | Interrupt Priority 2       |
| ICDIPR3  | 32   | 0           | rw | Interrupt Priority 3       |
| ICDIPR4  | 32   | 0           | rw | Interrupt Priority 4       |
| ICDIPR5  | 32   | 0           | rw | Interrupt Priority 5       |
| ICDIPR6  | 32   | 0           | rw | Interrupt Priority 6       |

|            |    |         |    |                                |
|------------|----|---------|----|--------------------------------|
| ICDIPR7    | 32 | 0       | rw | Interrupt Priority 7           |
| ICDIPR8    | 32 | 0       | rw | Interrupt Priority 8           |
| ICDIPR9    | 32 | 0       | rw | Interrupt Priority 9           |
| ICDIPR10   | 32 | 0       | rw | Interrupt Priority 10          |
| ICDIPR11   | 32 | 0       | rw | Interrupt Priority 11          |
| ICDIPR12   | 32 | 0       | rw | Interrupt Priority 12          |
| ICDIPR13   | 32 | 0       | rw | Interrupt Priority 13          |
| ICDIPR14   | 32 | 0       | rw | Interrupt Priority 14          |
| ICDIPR15   | 32 | 0       | rw | Interrupt Priority 15          |
| ICDIPR16   | 32 | 0       | rw | Interrupt Priority 16          |
| ICDIPR17   | 32 | 0       | rw | Interrupt Priority 17          |
| ICDIPR18   | 32 | 0       | rw | Interrupt Priority 18          |
| ICDIPR19   | 32 | 0       | rw | Interrupt Priority 19          |
| ICDIPR20   | 32 | 0       | rw | Interrupt Priority 20          |
| ICDIPR21   | 32 | 0       | rw | Interrupt Priority 21          |
| ICDIPR22   | 32 | 0       | rw | Interrupt Priority 22          |
| ICDIPR23   | 32 | 0       | rw | Interrupt Priority 23          |
| ICDIPTTR0  | 32 | 1010101 | rw | Interrupt Processor Targets 0  |
| ICDIPTTR1  | 32 | 1010101 | rw | Interrupt Processor Targets 1  |
| ICDIPTTR2  | 32 | 1010101 | rw | Interrupt Processor Targets 2  |
| ICDIPTTR3  | 32 | 1010101 | rw | Interrupt Processor Targets 3  |
| ICDIPTTR4  | 32 | 0       | rw | Interrupt Processor Targets 4  |
| ICDIPTTR5  | 32 | 0       | rw | Interrupt Processor Targets 5  |
| ICDIPTTR6  | 32 | 1000000 | rw | Interrupt Processor Targets 6  |
| ICDIPTTR7  | 32 | 1010101 | rw | Interrupt Processor Targets 7  |
| ICDIPTTR8  | 32 | 0       | rw | Interrupt Processor Targets 8  |
| ICDIPTTR9  | 32 | 0       | rw | Interrupt Processor Targets 9  |
| ICDIPTTR10 | 32 | 0       | rw | Interrupt Processor Targets 10 |
| ICDIPTTR11 | 32 | 0       | rw | Interrupt Processor Targets 11 |
| ICDIPTTR12 | 32 | 0       | rw | Interrupt Processor Targets 12 |
| ICDIPTTR13 | 32 | 0       | rw | Interrupt Processor Targets 13 |
| ICDIPTTR14 | 32 | 0       | rw | Interrupt Processor Targets 14 |
| ICDIPTTR15 | 32 | 0       | rw | Interrupt Processor Targets 15 |
| ICDIPTTR16 | 32 | 0       | rw | Interrupt Processor Targets 16 |
| ICDIPTTR17 | 32 | 0       | rw | Interrupt Processor Targets 17 |
| ICDIPTTR18 | 32 | 0       | rw | Interrupt Processor Targets 18 |
| ICDIPTTR19 | 32 | 0       | rw | Interrupt Processor Targets 19 |
| ICDIPTTR20 | 32 | 0       | rw | Interrupt Processor Targets 20 |
| ICDIPTTR21 | 32 | 0       | rw | Interrupt Processor Targets 21 |
| ICDIPTTR22 | 32 | 0       | rw | Interrupt Processor Targets 22 |
| ICDIPTTR23 | 32 | 0       | rw | Interrupt Processor Targets 23 |
| ICDISER0   | 32 | fff     | rw | Interrupt Set-Enable 0         |
| ICDISER1   | 32 | 0       | rw | Interrupt Set-Enable 1         |
| ICDISER2   | 32 | 0       | rw | Interrupt Set-Enable 2         |
| ICDISPR0   | 32 | 0       | rw | Interrupt Set-Pending 0        |
| ICDISPR1   | 32 | 0       | rw | Interrupt Set-Pending 1        |
| ICDISPR2   | 32 | 0       | rw | Interrupt Set-Pending 2        |
| ICDSGIR    | 32 | 0       | -w | Software-Generated Interrupt   |
| ICPIDR0    | 32 | 18      | r- | Peripheral ID 0                |
| ICPIDR1    | 32 | bc      | r- | Peripheral ID 1                |
| ICPIDR2    | 32 | b       | r- | Peripheral ID 2                |
| ICPIDR3    | 32 | 0       | r- | Peripheral ID 3                |
| ICPIDR4    | 32 | 4       | r- | Peripheral ID 4                |
| ICPIDR5    | 32 | 0       | r- | Peripheral ID 5                |
| ICPIDR6    | 32 | 0       | r- | Peripheral ID 6                |
| ICPIDR7    | 32 | 0       | r- | Peripheral ID 7                |

Table 13.14: Registers at level 2, type:CPU group:MPCore\_distributor

**13.2.15 MPCore\_processor\_interface**

Registers at level:2, type:CPU group:MPCore\_processor\_interface

| Name    | Bits | Initial-Hex | RW | Description                        |
|---------|------|-------------|----|------------------------------------|
| ICCBPR  | 32   | 0           | rw | Binary Point                       |
| ICCEOIR | 32   | 0           | -w | End of Interrupt                   |
| ICCHPIR | 32   | 3ff         | r- | Highest Priority Pending Interrupt |
| ICCIAR  | 32   | 3ff         | r- | Interrupt Acknowledge              |
| ICCICR  | 32   | 0           | rw | CPU Interface Control              |
| ICCIIDR | 32   | 3901043b    | r- | CPU Interface ID                   |
| ICCPMR  | 32   | 0           | rw | Interrupt Priority Mask            |
| ICCRPR  | 32   | ff          | r- | Running Priority                   |

Table 13.15: Registers at level 2, type:CPU group:MPCore\_processor\_interface

**13.2.16 MPCore\_timer\_and\_watchdog**

Registers at level:2, type:CPU group:MPCore\_timer\_and\_watchdog

| Name              | Bits | Initial-Hex | RW | Description                     |
|-------------------|------|-------------|----|---------------------------------|
| PTControl         | 32   | 0           | rw | Private Timer Control           |
| PTCounter         | 32   | 0           | rw | Private Timer Counter           |
| PTInterruptStatus | 32   | 0           | rw | Private Timer Interrupt Status  |
| PTLoad            | 32   | 0           | rw | Private Timer Load              |
| WTControl         | 32   | 0           | rw | Watchdog Timer Control          |
| WTCounter         | 32   | 0           | rw | Watchdog Timer Counter          |
| WTDisable         | 32   | 0           | -w | Watchdog Timer Disable          |
| WTInterruptStatus | 32   | 0           | rw | Watchdog Timer Interrupt Status |
| WTLload           | 32   | 0           | rw | Watchdog Timer Load             |
| WTRResetStatus    | 32   | 0           | rw | Watchdog Timer Reset Status     |

Table 13.16: Registers at level 2, type:CPU group:MPCore\_timer\_and\_watchdog

**13.2.17 MPCore\_global\_timer**

Registers at level:2, type:CPU group:MPCore\_global\_timer

| Name              | Bits | Initial-Hex | RW | Description                   |
|-------------------|------|-------------|----|-------------------------------|
| GTAutoIncrement   | 32   | 0           | rw | Global Timer Auto-Increment   |
| GTComparator      | 64   | 0           | rw | Global Timer Comparator       |
| GTControl         | 32   | 0           | rw | Global Timer Control          |
| GTCounter         | 64   | 0           | rw | Global Timer Counter          |
| GTInterruptStatus | 32   | 0           | rw | Global Timer Interrupt Status |

Table 13.17: Registers at level 2, type:CPU group:MPCore\_global\_timer